

Trainings ▾

Preparatory Steps

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Prepare the engine to remove the gear cover spacer plate.

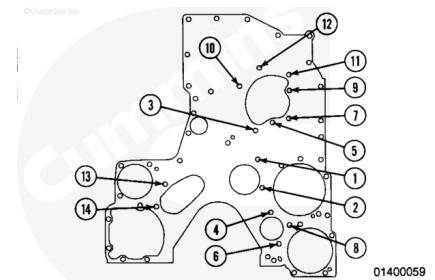
- Position the Number 1 cylinder on the compression stroke. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-012-tr.html>)
- Disconnect the battery. See equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>)
- Remove the water pump idler gear. Refer to Procedure 001-040 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-040-tr.html>)

- Remove the accessory drive. Refer to Procedure 009-001 in Section 9. (</qs3/pubsys2/xml/en/procedures/20/20-009-001-tr.html>)
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html>)
- Remove the camshaft gear. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-012-tr.html>)
- Remove the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-039-tr.html>)
- Remove the engine crankshaft speed/position sensor. Refer to Procedure 019-042 in Section 19, (</qs3/pubsys2/xml/en/procedures/19/19-019-042.html>) in the QSK19 CM850 Modular Comman Rail System Series Engine Troubleshooting and Repair Manual, Bulletin 4021493.

Remove

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Remove the front gear cover spacer plate.

Remove the 14 mounting capscrews from the plate.

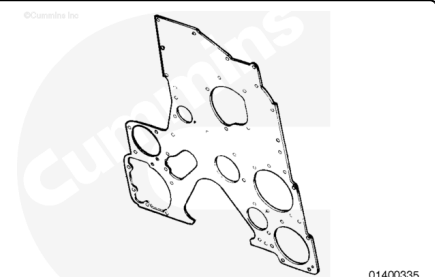
Remove the plate and discard the gasket.

Note : Tag the special mounting capscrews for future identification.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the gear cover spacer plate. Use solvent.

Dry with compressed air.

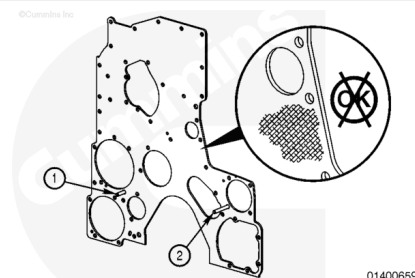
Inspect the plate for reuse. Check for fretting damage on both sides of the gasket surface. Replace the plate if damaged.

Check the mounting stud threads for damage.

Install the new studs to the specified height.

Use Loctite™ 577, Part Number 4919202, or equivalent.

Replace the studs if damage is found.



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Accessory Drive Stud (1) - Installed Height

mm		in
31.88	MIN	1.255
33.15	MAX	1.305

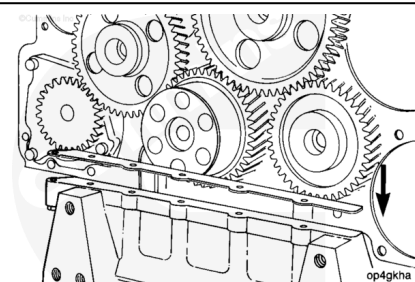
Water Pump Drive Stud (2) - Installed Height

mm		in
62.99	MIN	2.480
63.75	MAX	2.510

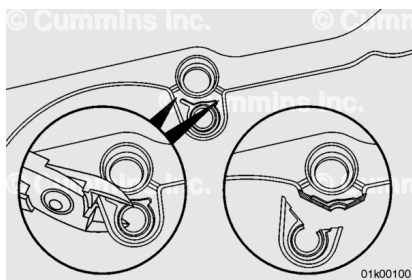
Install

Note : Remove the old pan adapter gasket at perforations. Gasket is designed to be removed at perforation. Steel core gasket can not be cut without causing damage to pan adapter surface.

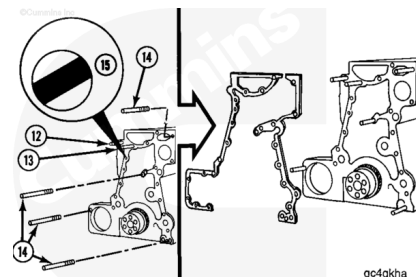
Use Cummins® silver silicon sealant, Part Number 3823494, at each of the four T-joints that intersect the oil pan gasket.



Remove the extraneous tab on the gasket by cutting along the two notched lines.



The diamond dowel (15) **must** be installed with the flat surface turned toward the master dowel hole at the lower right hand corner of the block. Guide studs (14) will aid the installation.

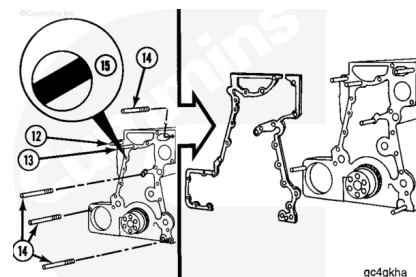


⚠ CAUTION ⚠

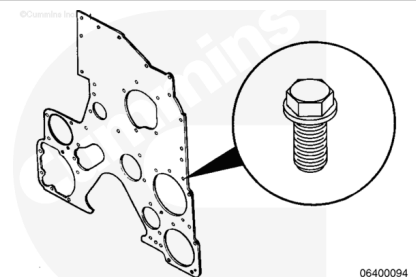
Do not use gasket cement. Damage to the gasket will result.

Install the gasket.

Note : If using edge molded rubber gasket, the bottom of the gasket will not be even with the bottom of the block. Fill this gap with sealant.

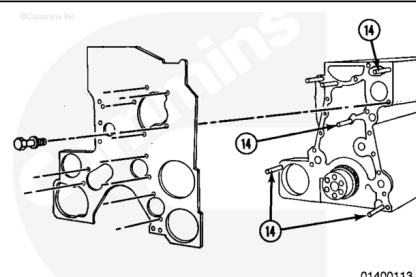


Note : Special capscrews are required to attach the spacer plate and the gear housing. The capscrews have a captive, cone-shaped washer to maintain torque.



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Install the spacer plate and the 11 capscrews as shown.
Remove the guide studs (14).

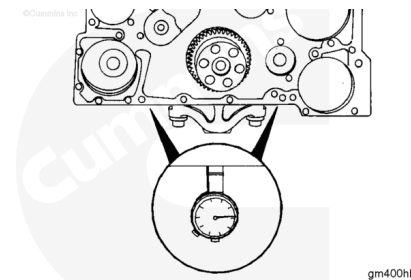
Install the remaining three capscrews.

Note : If the oil pan adapter is removed, spacer plate alignment **must** be checked. If the oil pan adapter is installed, alignment **does not** require checking.

Use a gauge block, Part Number ST-547, or equivalent, and measure the distance from the bottom of the block to the bottom of the plate.

The bottom of the plate or housing **must** be within 0.05 mm [0.002 inch] of the bottom of the block.

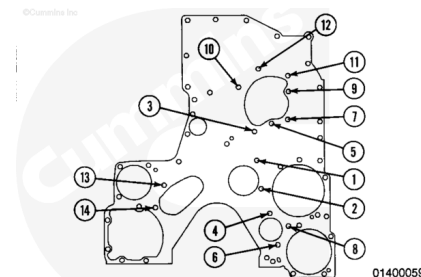
Align the plate with the bottom of the block, if necessary.



Tighten the capscrews. Use the sequence shown.

Torque Value: 45 N•m [33 ft-lb]

Measure the plate or the housing to the block alignment again to be sure it is within specifications.



Finishing Steps

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- Install the camshaft gear. Refer to Procedure 001-012 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-012-tr.html)
- Install the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-039-tr.html)
- Install the water pump idler gear. Refer to Procedure 001-040 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-040-tr.html)

- Install the camshaft idler. Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html)
- If the camshaft or camshaft gear was replaced, check the timing. Machining tolerances can cause the timing to be out of specifications. Refer to Procedure 006-025 in Section 6.
(/qs3/pubsys2/xml/en/procedures/20/20-006-025-tr.html)
- Install the accessory drive plate. Refer to Procedure 009-001 in Section 9.
(/qs3/pubsys2/xml/en/procedures/20/20-009-001-tr.html)
- Install the engine crankshaft speed/position sensor. Refer to Procedure 019-042 in Section 19, (/qs3/pubsys2/xml/en/procedures/19/19-019-042.html) in the QSK19 CM850 Modular Common Rail System Series Engine Troubleshooting and Repair Manual, Bulletin 4021493.
- Install the gear cover and related components. Refer to Procedure 001-031 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)
- Fill with lubricating oil. Refer to Procedure 007-037 in Section 7
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html).
- Fill with coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Connect the battery. See equipment manufacturer service information.
- Operate the engine. Check for leaks.